

Harvard-MIT Math Tournament February Guts 2017 — P9

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Jeffrey writes the numbers 1 and 10^8 on the blackboard. Every minute, if x, y are on the board, Jeffery replaces them with

$$\frac{x+y}{2} \quad \text{and} \quad 2\left(\frac{1}{x} + \frac{1}{y}\right)^{-1}$$

After 2017 minutes the two numbers are a and b . Find $\min(a, b)$ to the nearest integer.

Solution

Observe that

$$\frac{x+y}{2} \cdot 2\left(\frac{1}{x} + \frac{1}{y}\right)^{-1} = \frac{x+y}{2} \cdot \frac{2xy}{x+y} = xy$$

and so under the given operation the product of the two numbers on the board is invariant. Suppose that $y \geq x$. Then

$$\begin{aligned} \frac{x+y}{2} - \frac{2xy}{x+y} &< \frac{y-x}{2} \\ \frac{(y-x)^2}{2(x+y)} &< \frac{y-x}{2} \\ \frac{y-x}{x+y} &< 1 \end{aligned}$$

but this is true since $y - x < x + y$. Thus the gap between the two numbers on the board reduces by more than half each minute. Formally let D_k be the non-negative difference between the two numbers on the board after the k -th minute. Now since $\log_2(10^8) = 8 \log_2(10) < 32$ notice that

$$D_{2017} < \frac{1}{2^{2017}} \cdot 10^8 < \frac{1}{2^{2017}} \cdot 2^{32}.$$

So for some $\epsilon < \frac{1}{2^{1985}}$ the last two numbers on the board are of the form $(k - \epsilon, k + \epsilon)$ and have product 10^8 . Clearly they are both just very close to $\sqrt{10^8} = 10^4$ and so the answer is 10^4 .